

RACHEL ZOLL

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EDUCATION

Harvard University, Cambridge, MA

11/2025

Ph.D., Engineering Sciences — Microrobotics Laboratory · NSF Graduate Research Fellow

Thesis: Millimeter-Scale Multi-Material Manufacturing for Biomedical Devices — Advisor: Prof. Robert J. Wood

University of California, Berkeley, Berkeley, CA

05/2019

M.S., Electrical Engineering & Computer Science — Focus: MEMS, Digital ASIC Design

University of California, Berkeley, Berkeley, CA

05/2018

B.S., Electrical Engineering & Computer Science — Coursework: 28nm SoC for IoT, Digital Design & Integrated Circuits, Analog Integrated Circuits, Digital Signal Processing, Control Systems, Computer Architecture, Electronic Medical Devices, Power Electronics

TECHNICAL SKILLS

Sensor Systems & Characterization:

Sensor integration (IMU, load cells, force/pressure transducers); sensor characterization (accuracy, noise, repeatability); module- and system-level performance characterization; algorithm development & validation for sensor signal-processing pipelines.

Instrumentation & Validation:

Oscilloscopes & parameter analyzers; logic analyzers; hardware-in-the-loop (HIL) validation & custom DAQ software; Altium, KiCad.

Circuit & ASIC Design:

Digital circuit design (Verilog, CHISEL, FPGA/ASIC tapeout); ASIC toolchain exposure (layout, synthesis, verification); PCB design for high-speed/differential-pair signal integrity; Cadence; LTSpice/SPIICE; Xilinx FPGA.

Firmware & Programming:

C, Python, MATLAB, Assembly, Git; firmware debugging (GDB); STM32/microcontrollers; digital signal processing; HW/SW co-optimization for embedded sensor systems.

Protocols & Controls:

I2C, SPI, UART, CAN; PID & feedback control.

Design & Fabrication:

DFM/DFX; CAD (SolidWorks, Onshape); tolerance analysis; soft elastomer fabrication & molding; machining; 3D printing.

RESEARCH EXPERIENCE

Harvard University — Graduate Student Researcher, Microrobotics Lab

08/2019 – 10/2025

Advisor: Prof. Robert J. Wood

- Built HIL benchtop characterization suites with custom DAQ software for multi-domain (pressure, flow, thermal-signature) validation of custom micro-machined devices.
- Designed a custom flex PCB for system integration of a 32-channel skin-surface sensor array with COTS cable assemblies; validated via impedance testing, electrophysiology testing, and initial ex vivo trials.
- Engineered, laser-micromachined, and characterized a spring-driven mechanism for autonomous GI fluid-sampling capsules.
- Developed a soft-elastomer adhesion mechanism for fastener-free anchoring to soft tissue, validated via Instron testing.

UC Berkeley — Graduate & Undergraduate Researcher, Pister, Muller & Maharbiz Labs

08/2017 – 05/2019

- Designed and FPGA-verified a GFSK modulator, CDR, and image-reject filter as part of a team ASIC tapeout of a 28nm FDSOI Bluetooth transceiver ("Tapeout Class").
- Developed and verified a GFSK modulator (FPGA & ASIC) for a crystal-free 2.4GHz single-chip radio (SCuM).
- Optimized ARM Thumb assembly and modeled a MEMS finite-state machine for planar silicon motor operation.
- Simulated, fabricated, and tested MEMS actuators for parallel insertion of carbon-fiber neural recording electrodes.

PROFESSIONAL EXPERIENCE

Harvard Office of Technology Development & ARPA-Health — Research Analyst, Cambridge, MA

06/2023 – 04/2026

- Supported technical due diligence and commercialization strategy for 15+ hardware/deep-tech innovations across a \$50M+ portfolio, resulting in 2 industry licensing leads.
- Collaborated with Harvard OTD's Business Development team to evaluate commercialization potential and prepare an investor/industry-facing pitch deck for a Harvard technology, supporting its licensing and go-to-market strategy.

MGH & BWH Hospitals — Robotics Hardware Lead, MESH Innovation Teams

02/2020 – 06/2024

- Designed and deployed a hardware-in-the-loop (HIL) pressure-measurement validation platform used to qualify 3 device iterations; wrote embedded firmware for sensor readout and characterized measurement accuracy (± 1 –5%) and repeatability to establish a process-control standard for device validation.

- Led end-to-end industrialization of an autonomous feeding tube device: defined hardware requirements, managed a 5-person cross-functional team, and executed 2 in vivo swine trials to establish the first gold-standard placement-accuracy benchmark for a procedure with no prior clinical reference data.
- Applied DFM/DFX principles across 5+ design iterations, reducing part count by 50% and kit assembly time from ~20 minutes to ~5 minutes.

Facebook — Operations & Manufacturing Intern, Menlo Park, CA

05/2017 – 08/2017

- Designed and tested hardware interfaces for rapid diagnostic testing of server-rack power units.

Intel Corporation — SSD Failure Analysis Intern, Folsom, CA

05/2016 – 08/2016

- Automated electrical failure-analysis processes and improved debug workflows for PCIe storage products; trained colleagues in semiconductor wafer processing and MOSFET functionality.

FMC Schilling Robotics — Electrical Engineering Intern, Davis, CA

05/2015 – 08/2015

- Tuned PID feedback control for a heavy-duty underwater ROV hydraulic valve pack operating at high pressure, minimizing overshoot to prevent valve overload during subsea field deployment; drafted H-bridge microprocessor architecture for wireless power on hydraulic manipulators.

SELECTED PUBLICATIONS

- **R. Zoll**, M. Karpelson, S. Srinivasan, R.J. Wood, “Modular Ingestible Capsule Robot for Gastrointestinal Fluid Sampling,” *Advanced Functional Materials* (2025, in preparation).
- G. Maquignaz, **R. Zoll**, M. Karpelson, J.C. Weaver, R.J. Wood, “Design and fabrication of a parasite-inspired, millimeter-scale tissue anchoring mechanism,” *PNAS Nexus* (2024).
- E. Challita, T. Chen, **R. Zoll**, M. Yuen, R. Wood, “Osmosis-Driven Large-Scale Actuation for Shape-Shifting Mechanisms,” *IROS* (2025).
- F. Maksimovic, B. Wheeler, D. Burnett, O. Khan, S. Mesri, I. Suci, L. Lee, A. Moreno, **R. Zoll**, et al., “A Crystal-Free Single-Chip Micro Mote with Integrated sub-mW 802.15.4 Compatible Transceiver, BLE Compatible Beacon Transmitter, and Cortex-M0,” 2019 Symposium on VLSI Circuits, Kyoto, Japan, pp. C88–C89.
- **R. Zoll**, C.B. Schindler, T.L. Massey, D.S. Drew, M.M. Maharbiz, K.S.J. Pister, “MEMS-Actuated Carbon Fiber Microelectrode for Neural Recording,” *IEEE Transactions on NanoBioscience*, 8(2), pp. 234–239 (2019).
- D. Burnett, B. Kilberg, **R. Zoll**, O. Khan, K.S.J. Pister, “Tapeout Class: Taking Students from Schematic to Silicon in One Semester,” *IEEE International Symposium on Circuits and Systems (ISCAS)* (2018).

Full publication list of 12+ peer-reviewed papers available on request.

HONORS & FELLOWSHIPS

- NSF Graduate Research Fellowship, \$53,000/yr (2019 – 2022)
- HealthTech Fellowship Finalist, Harvard Medical School (2023)
- Certificate of Distinction in Teaching, Harvard Bok Center for Teaching and Learning (2020)
- EECS Departmental Fellowship & EECS Excellence Award, UC Berkeley (2018)
- Regent’s and Chancellor’s Scholarship, UC Berkeley, \$25,000/yr tuition fellowship (2014 – 2018)

TEACHING & LEADERSHIP

- Teaching Fellow, GENED 1080: Engineering the Acoustical World, Harvard University (Fall 2020, Fall 2023)
- Discussion TA, EE16A: Designing Information Devices and Systems I, UC Berkeley (Spring & Fall 2017)
- Lab Assistant, EE16A: Designing Information Devices and Systems I, UC Berkeley (Fall 2015)
- Harvard SEAS Graduate Council, Co-President (2020 – 2022); MIT Sidney Pacific Graduate Dorm Executive Council & Trustee (2021 – 2023)

LANGUAGES

Mandarin Chinese (Intermediate); French (Intermediate).